

When Do LLM-Generated Note Cards Help Retrieval and Reranking?

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Abstract

We study whether LLM-generated structured note cards, six-field summaries that compress each document into retrieval-oriented content, improve downstream search when introduced as an offline document representation. We first describe a training-free pipeline that turns each noisy Xiaohongshu (RedNote) post into a structured card, and we evaluate the reliability of two card-generation LLMs (generators) of different scales: a 4B generator produces well-formed cards on all but 0.01% of inputs, against roughly 4% for a 3B generator, so we adopt the 4B cards downstream. We then measure the effect of three document representations (the original text, the note card alone, and StructuredConcat, a structured concatenation of card and text) on the Qilin benchmark under expanded-pool retrieval and candidate-pool reranking. The effect is model-dependent and stage-dependent. For a BERT bi-encoder, the card helps, raising fine-tuned MRR@10 from 0.2748 to 0.3085 over the original-text baseline, while Qwen3-Embedding-0.6B gains little. In reranking, Qwen3-Reranker fine-tuned and evaluated on StructuredConcat reaches 0.6970 MRR@10, against 0.5260 for the Original-trained baseline, and most of this gain persists when inference uses only the original text, indicating a training-time rather than an inference-time effect. Within this benchmark and model set, our results suggest that the card is more reliable as an augmentation than as a replacement.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; *Search engine indexing*.

Keywords

document compression, synthetic content, dense retrieval, reranking, LLM-generated summaries

1 Introduction

Retrieval systems increasingly incorporate LLMs into search pipelines [21]. They also operate in ecosystems where synthetic and human-authored content coexist [11]. This raises a practical question: when an LLM-generated document representation is introduced into a retrieval pipeline, does it actually help? A compact, structured summary may surface query-relevant cues that a weak retriever would miss, but it may also discard distinctions that a stronger model can exploit from the original text. Work on source bias [3] shows that neural retrievers can prefer LLM-generated text, but studies bias detection rather than whether deliberately engineered synthetic representations improve ranking quality.

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Table 1: A representative raw post and its generated note card (English translation from Chinese).

Raw original post	
Title	Wuyuan Tourism Map (Latest 2024 Edition, Exclusively Drawn)
Body	The 2024 Wuyuan tourism map is out, including hidden attractions. Photos taken today (20240408) show Wuyuan's green village scenery. Do you find it beautiful? Tags: #Wuyuan hidden spots# #Wuyuan tourism# #latest Wuyuan map# #travel guide#
Generated note card	
Category	Travel
Keywords	Wuyuan tourism, tourism map, hidden attractions, latest 2024 edition, full-region tourism
Scene	Looking for the latest Wuyuan tourism map
Audience	Travel enthusiasts, Wuyuan visitors
Entities	Wuyuan
Gist	2024 latest Wuyuan tourism map including hidden attractions, with real-scene reference photos

We approach this question in two stages. First, we treat note card generation as an object of study: a training-free pipeline compresses each document into a structured *note card* with six retrieval-oriented fields (category, keywords, search scenario, target audience, named entities, and a one-sentence gist), and we evaluate how reliably generators of different scales produce well-formed cards. Second, we measure the downstream effect of these cards in a two-stage retrieve-then-rerank pipeline on Qilin [1], a large-scale search benchmark built from the Xiaohongshu (RedNote) platform, where documents are long, noisy, and heterogeneous [19]. Figure 1 summarizes the pipeline, and Table 1 shows a representative post and its generated card.

Our contributions are twofold: (1) a training-free structured-summary generation pipeline for noisy social-platform content, with a reliability analysis at scale; and (2) evidence that a fixed generative compression intervention acts differently across model classes and pipeline stages, improving a weaker BERT bi-encoder through the inference-time document view, leaving Qwen3-Embedding-0.6B with marginal gains, and improving Qwen3-Reranker mainly through card-augmented training. In this setting, the card proves more reliable as augmentation than as replacement.

2 Related Work

Retrieve-then-rerank pipelines. Modern search systems commonly pair a bi-encoder that selects candidates via dense scoring [10] with a cross-encoder that refines the ranking through joint query-document attention [13, 14, 16]; recent pretrained embedding models such as the Qwen3 Embedding series [20] provide strong out-of-the-box first-stage retrieval.

Document transformation for retrieval. Long and noisy documents pose challenges for neural ranking, motivating methods

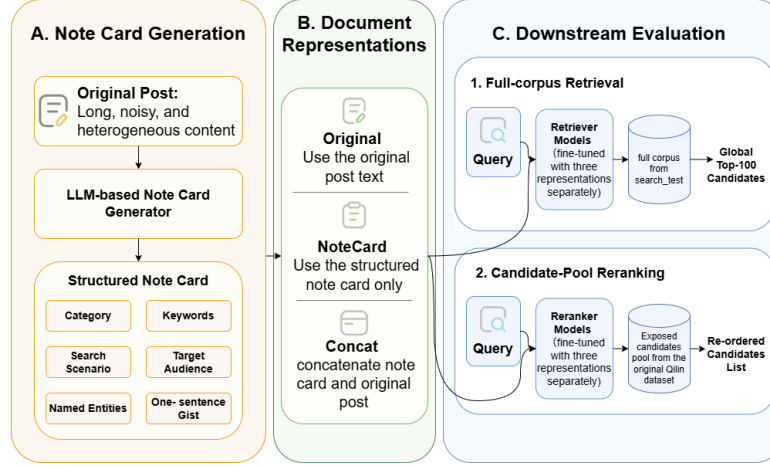


Figure 1: Overview of the pipeline. An LLM generates a structured note card from each post offline; the three document representations are then evaluated in global first-stage retrieval over an expanded corpus and in reranking of a query-specific candidate pool.

that transform the document representation rather than the ranking model alone. One line compresses: Gao and Callan [5] show that encoding long documents into low-dimensional representations creates an information bottleneck, and Jin et al. [9] use a fine-tuned LLM to decompose documents into tree representations compressed at query time. A complementary line synthesizes text: doc2query appends model-generated queries to each document before indexing [15], and HyDE generates a hypothetical document from the query at search time [6]. Our note card belongs to this generative line but is an offline, fixed-schema rewrite on the document side.

LLM-generated note representations. NoteLLM [19] compresses each Xiaohongshu note into a single latent embedding via a special token, trained jointly with hashtag generation for item-to-item recommendation. We instead produce an explicit textual summary and study how it affects different ranking models.

Source bias in neural retrieval. Dai et al. [3] show that neural retrievers tend to rank LLM-generated documents above human-written documents with equivalent content, a phenomenon they term *source bias*. Rather than replacing human-written documents, we prepend or substitute a structured summary and measure whether this helps or hurts ranking quality across model configurations.

3 Note Card Generation

Card schema. Given a note with title and body, an LLM generates a structured note card with six fields: category, keywords, search scenario, target audience, named entities, and a one-sentence gist. The keywords field mixes category, attribute, and intent terms a user might type; the scenario field gives a brief Chinese phrase describing the situation in which a user would issue such a search; the entities field extracts only brands, products, people, and places that appear explicitly in the source, with an explicit “none” marker

otherwise; and the gist is rewritten in plain declarative language with slang and marketing terms removed.

Generation procedure. Generation is training-free. A regular-expression denoising pass removes platform emoticon markers, emoji, redundant whitespace, and generic or low-information hashtags, while retaining potentially informative location, event, and brand tags as input evidence. The cleaned text is then placed into a prompt with four components: definitions of the six output fields, field-level constraints, five author-written exemplars, and an instruction to emit only the six labeled fields in a fixed line-oriented format. The exemplars cover fashion and outfit recommendations, automotive purchase advice, fan culture and concert logistics, regional food discovery, and consumer electronics troubleshooting; they were fixed before downstream evaluation and were not drawn from any Qilin query, test document, or relevance label, so the prompt does not leak test content. The exemplar set may nonetheless bias card style toward the covered categories. Both generators use the identical prompt and greedy decoding, so the comparison in Section 4 varies only the generator. The pipeline validates each card against the required field labels, so malformed or empty outputs can be regenerated.

Generators. We compare two open-source instruction-tuned generators of different scales, Qwen2.5-3B-Instruct [17] and Qwen3-4B-Instruct [18], and use the comparison (Section 4) to select the card set for all downstream experiments.

4 Generation Quality Evaluation

Malformed cards would confound downstream comparisons, so we first audit generation reliability on 22,598 cards from the Qilin test exposure pool. Qwen3-4B is substantially more reliable in format control: only 0.01% of its outputs fail to parse into the required six fields, compared with 4.04% for Qwen2.5-3B. Its per-field validity is

also above 99% for category, keywords, scenario, audience, and gist. The exception is named entities (46.54% valid, versus 43.10% for Qwen2.5-3B), mainly because many source posts contain no explicit brand, product, person, or place; these cases produce an explicit “none” marker and are counted as invalid by the strict check. We use the Qwen3-4B cards in all downstream experiments.

5 Downstream Evaluation

5.1 Problem formulation

A post is a pair $d = (t, b)$ of title t and body b . A denoising map ρ removes platform markup, and a generator g maps the cleaned post to a six-field card $c = g(\rho(d))$. Writing $\oplus$ for text concatenation and $\sigma(\cdot)$ for the serialization of the card into a single string, the three document representations are

$$x_O = t \oplus b, \quad x_N = \sigma(c), \quad x_S = \sigma(c) \oplus t \oplus b. \quad (1)$$

Original (x_O) keeps only the human-authored text, **NoteCard** (x_N) replaces it with synthetic content, and **StructuredConcat** (x_S) augments the text with the card, so the three trace a single replacement-to-augmentation axis. For a trained encoder we write $(r_{tr} \rightarrow r_{ev})$ for the configuration trained on $r_{tr} \in \{O, N, S\}$ and evaluated on r_{ev} . An off-diagonal pair such as $(O \rightarrow S)$ holds the trained weights fixed and varies only the inference-time representation; this is the comparison we rely on to separate a representation effect from a training effect, since the encoder is identical and only the document view changes.

Metrics. We report $MRR@k$, $MAP@k$, and $R@k$ in their standard definitions, with average precision normalized by $\min(k, R_q)$ where R_q is the number of relevant documents for query q . Retrieval applies these metrics over the expanded test-pool corpus; reranking applies them within each query’s candidate pool.

5.2 Setup

Dataset and splits. We use the Qilin search subset [1]: training on `search_train[0:4000]`, validation on `search_train[4000:5000]`, evaluation on `search_test[0:1000]`, scored with the official `search_test.qrels.csv` binary labels.

Document representations. All three representations are derived from the same Qwen3-4B card set and used unchanged in both protocols. For the cross-encoder rerankers we evaluate the full 3×3 train-inference grid; for the retrievers we evaluate the diagonal configurations together with the off-diagonal pairs that expose the replacement and augmentation effects most directly.

Protocols. We evaluate two protocols whose numbers are not directly comparable. *Expanded-pool retrieval* (Section 5.3) retrieves the global top-100 from a 113,548-note corpus formed from the union of the exposed candidates of `search_test[0:6192]`. *Candidate-pool reranking* (Section 5.4) reranks each query’s existing `search_results` candidates. We report $MRR@10$, $MAP@10$, and $R@10$ throughout.

Training and selection. The BERT bi-encoder is fully fine-tuned and scores documents by cosine similarity; its learning rate is selected on validation (best 5e-5). Qwen3-Embedding-0.6B is adapted with LoRA [7] on hard-mined negatives (best learning rate 2e-6)

and also evaluated zero-shot. The rerankers are trained on engagement labels under the same split. Every trained model is selected by validation $MRR@10$ and evaluated once on the held-out test split, so representation choices are never tuned on test data.

5.3 Note Cards in First-Stage Retrieval

We evaluate two neural retrievers: bert-base-chinese [4], zero-shot and after bi-encoder fine-tuning, and Qwen3-Embedding-0.6B [20], zero-shot and after LoRA fine-tuning. Table 2 reports the diagonal configurations together with the off-diagonal rows that isolate replacement, augmentation, and training-transfer effects.

Table 2: Expanded-pool retrieval over the 113,548-note test corpus. Trained rows show training representation $\rightarrow$ inference representation (O = Original, N = NoteCard, S = StructuredConcat). Bold marks the best value per block; the zero-shot Qwen3-Emb. block is left unbolded because Original and StructuredConcat are within 0.0003 of each other on every metric, a gap we treat as indistinguishable under single-seed evaluation (Section 5.3).

Retriever	Representation	MRR@10	MAP@10	R@10
BERT bi-enc. (zero-shot)	Original	0.0630	0.0213	0.0515
	NoteCard	0.2063	0.0907	0.1861
	StructuredConcat	0.1761	0.0731	0.1519
BERT bi-enc. (fine-tuned)	O $\rightarrow$ O	0.2748	0.1310	0.2637
	O $\rightarrow$ N	0.2878	0.1466	0.2950
	O $\rightarrow$ S	0.3085	0.1552	0.3022
	S $\rightarrow$ O	0.2000	0.0836	0.1748
	S $\rightarrow$ N	0.2725	0.1342	0.2713
	S $\rightarrow$ S	0.2937	0.1490	0.2952
Qwen3-Emb. (zero-shot)	Original	0.4187	0.2492	0.4719
	NoteCard	0.4004	0.2332	0.4508
	StructuredConcat	0.4188	0.2494	0.4722
Qwen3-Emb. (LoRA)	O $\rightarrow$ O	0.4194	0.2519	0.4722
	O $\rightarrow$ N	0.3959	0.2337	0.4506
	O $\rightarrow$ S	0.4151	0.2464	0.4721
	S $\rightarrow$ O	0.4205	0.2501	0.4729
	S $\rightarrow$ N	0.3963	0.2330	0.4506
	S $\rightarrow$ S	0.4145	0.2484	0.4736

The card mainly helps the weaker bi-encoder. In the zero-shot setting, replacing Original with NoteCard raises BERT $MRR@10$ from 0.0630 to 0.2063, suggesting that the card supplies surface-level search cues this encoder does not reliably extract from noisy posts. After fine-tuning, the most controlled comparison holds the Original-trained checkpoint fixed and varies only the inference representation: NoteCard raises $MRR@10$ from 0.2748 to 0.2878, and StructuredConcat to 0.3085. Because the checkpoint is unchanged, the improvement is consistent with an inference-time representation effect rather than an effect of training on synthetic text. Training directly on StructuredConcat is also effective (S $\rightarrow$ S, 0.2937) but lower than Original-trained inference-time augmentation, and the S $\rightarrow$ O transfer collapses to 0.2000, so a checkpoint trained on the augmented view does not recover when the card is removed. For this encoder, the card is more useful as an additional document view than as a replacement training signal.

The strong encoder shows the opposite inference-time direction. Qwen3-Embedding-0.6B behaves differently. Zero-shot, Original and StructuredConcat are nearly tied (0.4187 and 0.4188) and NoteCard-only is lower. The LoRA grid sharpens this: holding the Original-trained checkpoint fixed, switching inference to StructuredConcat slightly lowers MRR@10 (0.4194 to 0.4151) and NoteCard lowers it further (0.3959), the opposite of the inference-time direction observed for BERT. The best LoRA configuration (S→O, 0.4205) improves over zero-shot by only 0.0018 MRR@10, and given single-seed evaluation we treat differences of this magnitude as indistinguishable. The same card thus has limited marginal value once the retriever already performs well on the original posts.

5.4 Note Cards in Reranking

Reranking tests how representations affect ordering within each query’s exposed candidate pool. Because this pool already contains almost all positives (Recall@100 $\approx$ 0.99 for cross-encoders), MRR@10 and MAP@10 are more informative than recall. We fine-tune Qwen3-Reranker-0.6B, MacBERT [2], and RoBERTa [12] cross-encoders under the same split. Table 3 reports the full train-inference MRR@10 grid per family; MAP@10 follows a similar pattern except where noted.

Table 3: Candidate-pool reranking, MRR@10 for each train-inference representation (rows) evaluated on each inference representation (columns).

Reranker	Train	→ O	→ N	→ S
Qwen3-Reranker-0.6B	O	0.5260	0.5770	0.5219
	N	0.5951	0.6207	0.5896
	S	0.6784	0.5903	0.6970
MacBERT cross-enc.	O	0.5169	0.5072	0.5030
	N	0.5041	0.5072	0.5106
	S	0.4859	0.5077	0.5071
RoBERTa cross-enc.	O	0.4911	0.5086	0.4949
	N	0.4858	0.5032	0.4914
	S	0.4890	0.5052	0.4969

For Qwen3-Reranker, the gain is primarily a training effect. Qwen3-Reranker trained and evaluated on StructuredConcat reaches 0.6970 MRR@10 (MAP@10 0.4811), against 0.5260 (MAP@10 0.3767) for the Original diagonal. The full grid shows where this gain comes from: the StructuredConcat-trained checkpoint still reaches 0.6784 when inference uses only the original text, and on MAP@10 this configuration is even slightly ahead (0.4865 versus 0.4811), so most of the improvement is attributable to training on the card-augmented view rather than to seeing cards at inference. Training on NoteCard alone is also effective (0.6207 on its diagonal), and every card-trained configuration outperforms the Original diagonal. For this model, the training representation dominates the inference representation, the reverse of the BERT bi-encoder, where the largest gains came from changing the inference view under a fixed Original-trained checkpoint. The smaller cross-encoders behave differently again: for MacBERT, no card-based configuration beats the Original diagonal on MRR@10 (best 0.5106 for N→S against 0.5169), and training on StructuredConcat transfers poorly back to Original inference (0.4859); RoBERTa scores highest on NoteCard inference under every training representation, with O→N (0.5086) above its Original

diagonal (0.4911). In this setting, card-based training helps substantially for Qwen3-Reranker, while the smaller cross-encoders show weaker, representation-specific effects in both directions.

6 Analysis

Where synthetic content helps. The same note card produces different effects across models and stages, and the grids expose two distinct mechanisms. For the weak BERT bi-encoder, the card acts through the inference-time document view: gains appear when the card is added at inference under a fixed Original-trained checkpoint. For Qwen3-Reranker, the card acts through the training signal: gains from card-augmented training persist even under original-text inference, while MacBERT and RoBERTa show smaller, representation-specific changes. Two regularities are therefore supported within this benchmark and model set: augmentation is more reliable than replacement, and the value of synthetic document content depends on the downstream model and ranking stage. This qualifies the source-bias result of Dai et al. [3]: a preference for LLM-generated text does not by itself imply better ranking quality, consistent with recent evidence that such preference originates in dataset supervision artifacts rather than in the retriever architecture [8].

Limitations. The encoders differ in architecture, capacity, pre-training data, and retrieval-specific training at once, so the comparison establishes model dependence but cannot isolate its driver. The first-stage corpus is not the platform’s full index but an expanded test-pool corpus formed from the union of exposed candidates, which bounds the external validity of the retrieval numbers. We report point estimates from a single training seed without significance testing, so small differences should be read as indistinguishable. Training relevance comes from implicit engagement while test relevance uses the official binary qrels. Card quality is audited only for format validity rather than semantic faithfulness or coverage. Finally, the six-field schema was designed for Xiaohongshu content and may not transfer to other document types, and all downstream experiments use a single (4B) card set, leaving the effect of generator scale on downstream ranking unmeasured.

7 Conclusion

We studied LLM-generated structured note cards as both a generation problem and a downstream document representation. The card improves BERT retrieval through the inference-time document view and Qwen3-Reranker reranking through the training signal, but provides little benefit for Qwen3-Embedding-0.6B and inconsistent benefit for smaller cross-encoders, supporting a model- and stage-dependent view of synthetic document representations: generation quality is necessary, but downstream value depends on how the representation is used. Future work should test additional seeds, audit the semantic faithfulness and coverage of generated cards, and explore score fusion between original- and card-based rankings.

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